



Framework for Blockchain-Based Community-Assisted Privacy- Preserving Reporting Service for Local Governance

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Abstract

Local governments in developing countries often struggle to address public issues efficiently. These struggles are mainly due to limited transparency, centralized data management, and insufficient citizen participation. E-governance and crowd-sourced reporting systems are designed to enhance public engagement. But their effectiveness is limited by issues related to data integrity, user anonymity, content reliability, and trustworthiness. As a solution to the above challenges, we propose a blockchain-based, community-assisted, privacy-preserving reporting framework for local governance. The main objective of the project is to design and prototype a permissioned blockchain-based reporting system. This system will enable citizens to anonymously submit public issue reports and participate in opinion polling. Also, our system will ensure data immutability and transparency. To manage reporting workflows, issue tracking, and community voting in a secure and tamper-resistant manner, the blockchain will use Proof-of-Authority (PoA) mechanism and smart contracts. User identity is decoupled from the report content using a simulated identity and an access control mechanism to preserve the privacy of the users. Large data objects, such as images, are stored off-chain with cryptographic hashes anchored on the blockchain. Additionally, an Artificial Intelligence (AI)-assisted content moderation mechanism will be used to minimize the impact of malicious, low-quality reporting content. To demonstrate the feasibility of the proposed framework, A web-based Decentralized App (DApp) is developed as a proof of concept within a controlled experimental environment. Through this project, we aim to deliver a practical and scalable prototype that enhances citizen participation, accountability, and trust in local governance systems.

Acknowledgments

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Contents

Abstract	i
Acknowledgements	ii
Contents	v
List of Figures	vi
List of Tables	vii
Acronyms	viii
1 Introduction	1
1.1 Problem Background	1
1.2 Problem Statement	2
1.3 Objectives and Scope	3
1.3.1 Aim	3
1.3.2 Objectives	3
1.3.3 Scope	3
1.4 Literature Review	4
1.4.1 Digital Citizen Reporting and e-Governance Systems	4
1.4.2 Blockchain for Local Governance Applications	5
1.4.3 Blockchain-Based Citizen Reporting and Complaint Man- agement Systems	5
1.4.4 Permissioned Blockchains and PoA Consensus Mechanisms	7
1.4.5 Smart Contracts for Public Issue Management and Opinion Polling	8
1.4.6 User Interfaces for Blockchain-Based e-Gover- nance Applications	9
1.4.7 AI-Based Content Moderation in Civic Reporting	11
1.4.8 Need for Automated Content Moderation	11
1.4.9 Architectural Approaches for AI Blockchain Integration	11

1.4.10	AI Oracles and Decentralized Oracle Networks	12
1.5	Report Outline	13
2	Background	14
2.1	Blockchain Technology	14
2.2	Permissioned Blockchain and PoA	14
2.3	Smart Contracts	15
2.4	Decentralized Applications (DApps)	15
2.5	InterPlanetary File System (IPFS)	16
2.6	Zero-Knowledge and Identity Concepts	17
2.7	AI-based Content Moderation	17
3	Methodology and Implementation	18
3.1	Methodology	18
3.1.1	Research Methodology	18
3.1.2	System Requirements Analysis	19
3.1.3	Proposed System Architecture	20
3.1.4	Proposed Workflow	23
3.1.5	Project Plan and Timeline	23
3.2	Project Timeline	23
3.2.1	Expected Outcomes	23
3.3	Implementation	25
3.3.1	Permissioned Blockchain Implementation	25
3.3.2	Smart Contract Architecture and Logic	25
3.3.3	Identity and Access Control Implementation	28
3.3.4	Backend Relayer Implementation	29
3.3.5	IPFS Off-chain Storage Integration	31
3.3.6	AI-based Moderation System Implementation	32
3.3.7	Web-based DApp Development	33
4	Testing, Experimentation, and Validation	36
4.1	Experimental Setup	36
4.1.1	Hardware Infrastructure	36
4.1.2	Software and Deployment Environment	37
4.1.3	Blockchain Configuration (Server 1)	37
4.1.4	Application and Services Topology	37
4.2	Testing Methodology	38
4.2.1	Subsystem Testing	38
4.2.2	Integration Testing	39
4.2.3	Functional Testing	39
4.3	Blockchain Performance Testing	40

4.4	Smart Contract Validation	40
4.5	AI Moderation Testing	40
4.5.1	API Availability Test	41
4.5.2	Moderation Classification Tests	41
4.5.3	Oracle Voting Validation	42
4.5.4	Response Structure Validation	42
4.6	IPFS Storage Testing	43
4.7	End-to-End Integration Testing	44
4.8	Results and Observations	49
4.9	Comparison with Existing Systems	49
4.9.1	Centralized Reporting Platforms	49
4.9.2	Traditional Content Moderation	49
4.9.3	Public Blockchain Approaches	50
4.9.4	Domain-Specific Blockchain Solutions	50
5	Discussion	51
5.1	Novel Contributions of the Project	51
5.2	Overall System Validity	51
5.3	Advantages of Proposed Architecture	51
5.4	Limitations of Current Prototype	51
5.5	Real-world Applicability	52
6	Conclusion and Future Work	53
6.1	Conclusion	53
6.2	Future Work	53
A	Smart Contract Code	54
B	Oracle API Endpoints	55
C	Docker Compose Files	56
D	System Diagrams	57
E	Testing Screenshots	58
F	GitHub Repository Links	59
	Bibliography	60

List of Figures

1.1	System Architecture of the Police Complaint Management System.[?]	6
3.1	High Level Architecture Diagram.	21
3.2	Project Timeline Gantt Chart	24
3.3	State transition diagram for the civic reporting system.	27
3.4	Backend relay workflow architecture.	30
3.5	Oracle network architecture for AI moderation.	32
4.1	Citizen Authentication Interfaces	45
4.2	Report Submission Form	46
4.3	Views of the Submitted Report in the Community Feed	48

List of Tables

4.1	AI moderation API availability tests	41
4.2	AI moderation classification tests	42
4.3	Oracle voting validation	42
4.4	Oracle aggregator response fields	43

Acronyms

AC - Alternating Current
DC - Direct Current
LV - Low Voltage
MV - Medium Voltage

Chapter 1

Introduction

Local governance is fundamental to the upkeep of public infrastructure and the overall well-being of the community. To address public problems such as damaged roads, issues in waste management, street light failures, effective communication between citizens and local authorities is critical. Recently, the integration of e-governance systems and crowdsourcing has improve the citizen participation and streamline the reporting and resolution of public issues.

However, most of the current systems rely on centralized architectures. This poses a challenge in dealing with data integrity, transparency, and user privacy. Citizens are reluctant to provide complaints about sensitive topics or criticism due to potential retaliation or the possibility of their personal info being abused. At the same time, the local government is faced with the challenge of verifying the authenticity of anonymous complaints raised by citizens.

Some of the key features of Blockchain technology are immutability, transparency, and decentralization. Those features can be used to address several limitations of conventional reporting systems. It is possible to design reporting platforms that protect user anonymity while ensuring the integrity and accountability of reported information by combining blockchain-based architectures and privacy-preserving mechanisms. This project examines the application of blockchain technology in developing a secure, community-driven reporting framework tailored to local governance contexts.

1.1 Problem Background

In the context of local governance, crowdsourced reporting systems and e-governance platforms are widely adopted. By using those systems, citizens can report issues, provide feedback and participate in decision making. It improves the inclusiveness and the accountability of every citizen. In developing countries like Sri Lanka, these systems have the potential to significantly improve the governance efficiency.

Timely identification and the resolution of public concerns are the key to governance efficiency.

However, despite their advantages, the conventional e-governance and reporting systems are still highly centralized, making them prone to various threats such as data tampering, illegal access, and single points of failure. In addition, the ability to track behavior down to individual users could serve as a discouraging factor, particularly in reporting delicate matters, in a system that lacks anonymity. Blockchain-based systems have appeared as a feasible solution for managing shared data in a tamper-proof and transparent manner. The ability of permissioned blockchains to enable controlled participation while maintaining accountability through trusted validators makes them a desirable choice for governance applications. By using smart contracts, reporting workflows and opinion polling can be automated rather than relying on centralized intermediaries.

While blockchain technology offers data integrity and transparency, it still poses challenges related to user privacy, content moderation, and trustworthiness among anonymous participants. These challenges remain open research and implementation problems. Combining blockchain with privacy-preserving identity mechanisms and AI-assisted moderation can help address these limitations and supports the need for a comprehensive framework for secure and reliable community-assisted reporting.

1.2 Problem Statement

The existing frameworks of e-governance and crowdsourced reporting systems for local governance do not provide an integrated mechanism that can guarantee data integrity, user anonymity, and trustworthiness of citizen submitted reports. Centralized architectures are prone to unauthorized access, data tampering, and loss of transparency. At the same time, identity dependent systems discourage citizen engagement due to privacy concerns and fear of retaliation.

Although blockchain has been applied to the context of local government context, many blockchain-based reporting systems still fail to balance anonymity with mechanisms for assessing the reliability and quality of submitted reports. Furthermore, the lack of content moderation and trust evaluation increases the risk of misinformation, and malicious or low quality submissions.

Therefore, there is a clear need for a privacy-preserving, blockchain-based reporting framework that enables anonymous citizen participation while ensuring data immutability, transparent workflows, and reliable evaluation of reported information. This project aims to address this gap by designing and prototyping a permissioned blockchain-based community-assisted reporting system that incorporates smart contracts, off-chain data storage, and AI-assisted content moderation mechanism

suitable for local governance environments.

1.3 Objectives and Scope

1.3.1 Aim

The aim of this project is to design and prototype a blockchain-based, privacy-preserving, community-assisted reporting system for local governance that empowers citizens to report public issues and provide their opinions in a transparent, accountable, and secure manner.

1.3.2 Objectives

1. Design and implement a permissioned reporting blockchain using a PoA consensus mechanism to ensure the integrity and immutability of citizen-submitted reports.
2. Design and deploy smart contracts to automate public issue reporting workflows, including ticket creation, issue resolution and closure, community voting as feedback on resolved issues, and separate opinion polling mechanisms for public consultation.
3. Develop user-friendly interfaces that enable citizens and authorized government officials to interact with the blockchain for reporting issues and tracking their resolution status.
4. Design and integrate an AI-based content moderation system to detect and filter inappropriate textual and visual content in user-submitted reports.

1.3.3 Scope

1. A prototype of the proposed blockchain will be implemented with a minimum of three full nodes. The PoA consensus mechanism will be used to enable efficient and controlled transaction validation within a permissioned network, while excluding deployment of public, permissionless blockchains or globally distributed ledger infrastructures.
2. User authentication and identity decoupling are managed through a simulated identity service, and the design and implementation of a fully decentralized Self-Sovereign Identity (SSI) blockchain are explicitly excluded from the scope of this project.

3. The integration of AI within the system is limited to content moderation using existing machine learning models, libraries, or external Application Programming Interfaces (Application Programming Interface (API)s), and the development of custom, from-scratch deep learning architectures is not within the scope of this work.
4. Transaction validation authority within the blockchain network is restricted to pre-approved institutional stakeholders, including local government entities and recognized non-governmental organizations, and does not permit anonymous public participation or Proof-of-Work (Proof-of-Work (PoW)) based mining.
5. The project emphasizes functional validation, system evaluation, and performance analysis within a controlled experimental environment, and does not address real-world legal compliance, regulatory adoption, legislative reform, or large-scale integration with existing national government information systems.
6. The user interface development is limited to a responsive, web-based DApp accessible via standard mobile and desktop web browsers, and explicitly excludes the development of native mobile applications (iOS or Android) or cross-platform native frameworks.

1.4 Literature Review

1.4.1 Digital Citizen Reporting and e-Governance Systems

Electronic governance (e-governance) is using the information and communication technologies to provide government services. This includes the support of government to citizens (G2C), government to business (G2B) , government to agencies (G2A) etc. By utilizing the e-governance platforms, everyone can gain access to whole range of public services. Additionally citizens can submit complaints and participate in governance decision making process. [?].

Digital citizen reporting system is one of the many use case of e-governance. "FixMyStreet" is a a such existing platform. It enable citizens to report issues for different issue categories. Sanitation, Traffic and Public Safety are some examples. Previous studies shows that these kind of systems improve governance administrative efficiency, civic participation and also they increase the transparency.[?].

Even though there are many benefits of digital citizen reporting in local governance as described earlier, most of them still use centralized server based architecture.

This centralized approach have its own limitations and disadvantages. Susceptibility to data tempering, lack of transparency are the main problems that impact heavily on public trust. In addition to that, mandatory identity disclosure is discouraging public participation on this kind of reporting systems. Therefore these problems motivate the findings of new decentralized, transparent and immutable architectures that still preserve user privacy with accountability.

1.4.2 Blockchain for Local Governance Applications

Blockchain is a technology that has been popular for its ability to have an immutable, transparent and decentralized ledger for public data. Hence, its a transformative tool for e-governance to address the problems that existing implementations had. The Blockchain allow us to verify the immutability of the records, so its suitable for governance applications where the trust and the accountability is crucial. [?], [?].

Blockchain based land registry system implemented in Cook Country, Illinois is a one real world example of the applicability of blockchain in local governance. They aim to reduce the property fraud by consolidating historical ownership records into a verifiable block in the blockchain. Similar to that Delaware has taken initiative to explore the use of smart contracts that's enable real time asset tracking and automate business fillings. This improves the regulatory compliance and oversights [?].

In addition to administrative efficiency, blockchain has been utilized for citizen centric services, such as secure digital identity management. Estonia has taken a initiative to integrate blockchain across healthcare, taxation and even electronic voting systems. So this can provide a mathematically verifiable audit trail for government data. MyPass project in Austin and the World Food Program's building blocks initiative also use Blockchain technology under the hood to manage encrypted identity and transaction records [?].

Despite the advancements in the blockchain base applications, existing literature highlights the challenges of adoption in governance. Scalability constraints, privacy concerns related to identity data, performance overhead and also the need of the legislative and institutional reforms are some examples. These factors needs to be addressed by carefully selecting blockchain architectures and consensus mechanisms before implementing systems for frequent citizen interactions.

1.4.3 Blockchain-Based Citizen Reporting and Complaint Management Systems

The traditional centralized platforms have data integrity issues, transparency issues and trust issues when it comes to citizen reporting. Past works also indicates

that centralized complaint registries often have single point of failures , delayed responses and public mistrust. Therefore blockchain integrated reporting services can be used to mitigate those issues.

A blockchain based police complaint management system has proposed by Pratheeba [?] contains a modular architecture. It's include security, blockchain, web interface and a hybrid cloud storage. This system ensure immutability of complaint records by cryptographic hash linking. And also it utilize RSA based encryption to restrict the access only to authorized personnel. In Addition to that, Pratheeba used a Natural Language Processing (NLP) techniques to automatically categorize complaints in the system based on their priorities.

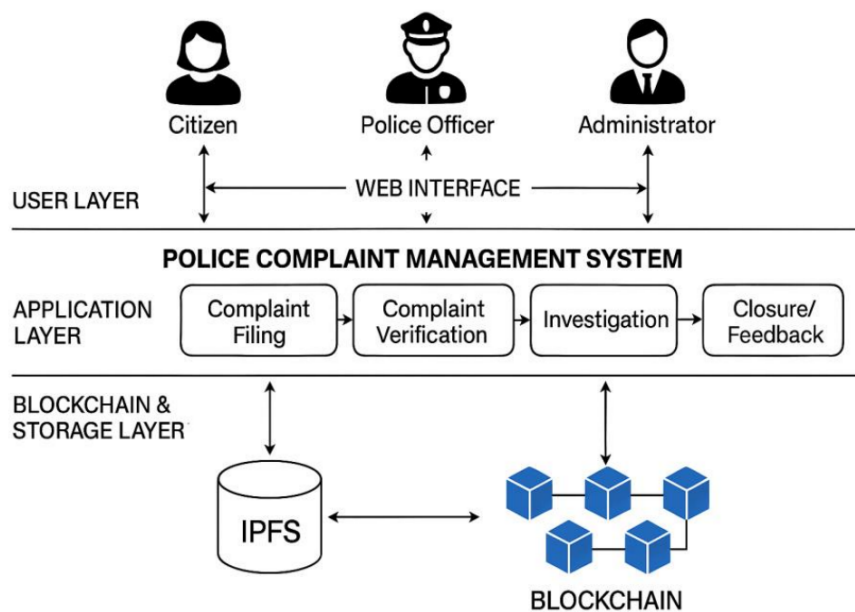


Figure 1.1: System Architecture of the Police Complaint Management System.[?]

A three tier architecture for decentralized complaint management framework has proposed by Manjula [?] is shown in the Figure 1.1. The three layers are web based presentation layer, a business logic layer and finally a blockchain data layer. This system has a role based access control method to involve multiple stakeholders. Citizens, Law enforcement authorities and judicial bodies are some of them. More importantly this system has a immutable storage for multimedia evidences. Intelligent complaint assignment based on workloads, automatic escalation mechanism are some of the key features of the framework. The experimental evaluations shows that a significant reductions in complaint processing time. The successful detection of data tampering also archived by blockchain verification.

In literature, existing blockchain based complaint management systems shows some improvements in transparency and administrative efficiency. But most of them are domain specific. They primarily target law enforcement or isolated governance functions. In addition to that several platforms still rely on centralized architectures. Only a limited attention is given for implementing a community assisted validation, for privacy preserving reporting mechanisms and efficient handling of multimedia evidences. Therefore these limitations highlight the need for a generalized reporting framework for local governance.

1.4.4 Permissioned Blockchains and PoA Consensus Mechanisms

Based on the participation and access control blockchain network can be classified in to two categories. [?]. They are permissionless architecture (public) and permissioned architecture. Bitcoin and Early Ethereum networks are permissionless blockchains. They allowed unrestricted participation and relied on consensus mechanism like PoW or Proof of Stake (Proof-of-Stake (PoS))[?]. These architectures and consensus mechanisms emphasized decentralization among participants. But they often suffer from the high latency of the network, limited throughput and unpredictable transaction costs due to congestion. These problems make the traditional permissionless PoW or PoS consensus mechanisms unsuitable for governance and crowdsourced applications[?].

While permissionless blockchains are unsuitable for governance related applications, permissioned blockchains restrict the network participation to authorized entities. This enables more efficient consensus mechanisms and greater control over governance data [?]. This known identities of participants allow permissioned systems to achieve higher performance. And support the regulatory compliances, data privacy and structured governance models as well [?]. Other than PoW and PoS consensus mechanisms, PoA offer significant advantages over permissioned blockchains. PoW requires energy intensive computation. PoS relies on the economic stake and introduce wealth centralization risks. But on the other hand PoA depends on the identity and the reputation of pre approved validators[?], [?]. Therefore this PoA consensus mechanism enables high transaction throughput and low latency. And also it mitigate the issues such as the "nothing at stake" problem by off chain legal and reputational accountability [?], [?].

The government systems should have accountability, performance predictability and data confidentiality. The PoA is well suited consensus mechanism for that [?]. By using PoA we can have the ability to identify and regulate validator entities to align blockchain operations. It's because we have the existing legal frameworks that ensure institutional responsibility. Hence, can avoid malicious behavior [?].

In Addition to that PoA avoids the indeterministic transaction fees and support controlled validation environments. This make them viable for handling sensitive governmental data under regulatory constraints[?],[?]. As a result, PoA emerges as a practical consensus mechanism for permissioned blockchain-based reporting systems in local governance contexts.

1.4.5 Smart Contracts for Public Issue Management and Opinion Polling

In governance systems, to improve the transparency, efficiency and citizen participation, smart contracts have been explored as the good mechanism. Smart contracts can automate the governance rules and incentives by immutable code. It can support models like civic intelligence governance (CIG). In CIG, citizen engagement is encouraged through on chain participation and token based incentives[?]. Literature shows that approaches like CIG can improve participation rates and knowledge sharing while reducing relances on centralized intermediaries. In the context of multi organizational governance smart contracts also been used within dual blockchain architectures. This method separates transaction data from contract logic. It balance the transparency and privacy in Government to Government and Business to Business processes[?].

Smart contracts also support business processes in governance by embedding policy rules directly into executable logic. Bid submission, evaluation and contract execution are some of the use cases in the public procurement and tendering processes that smart contract hold its value. It also reduce the oppertunities for corruption and increase the accountability[?], [?], [?]. Similar benefits are observed in the blockchain based land registry systems. Smart contracts have enable immutable property records automated ownership transfers and enforcement of land use policies. This has reduced fraud and administrative inefficiencies[?], [?].

In the context of voting and opinion polling, smart contracts offer enhanced security, transparency, immutability and verifiability inherited from blockchain[?], [?]. There are advance mechanisms such as Quadratic Voting and also blockchain enabled participatory budgeting. Those mechanisms further extend citizen involvement in decision making. It have expressive preference representation and transparent allocation of public resources[?], [?]. In addition to these advantages, there exists some challenges related to scalability, privacy and system complexity when deploying voting and polling systems to large and diverse populations[?], [?].

1.4.6 User Interfaces for Blockchain-Based e-Governance Applications

User interfaces play a critical role in determining the accessibility and adoption of blockchain based e-governance systems. In the traditional applications the user literacy is less important but in this conventional blockchain applications the user literacy is highly considered because of the blockchain concepts that are different from the traditional apps and it does make the users hard to understand and adopt. The literature consistently identifies usability and user experience challenges as major barriers to the adoption of decentralized applications for the general public. Particularly where security requirements conflict with established usability principles [1], [2]. So the design of intuitive, inclusive, and responsive web interfaces is essential for enabling effective interaction between citizens, government officials, and the underlying blockchain infrastructure.

Usability Challenges in Decentralized Governance Systems

Decentralized applications are fundamentally different from traditional web systems due to its architectural characteristics. Blockchain based systems have asynchronous transactions, immutable records, and cryptographic identity management. They can create friction for users who are used to centralized platforms. Studies highlight a significant usability gap in Web3 applications because of the concepts like transaction finality, gas fees, and private key ownership which are difficult for non technical users to understand and manage [1], [2].

Identity Management and Cognitive Load

One of the most critical usability barriers is wallet based identity management. The reason is user identity is tied to cryptographic key pairs instead of institution managed accounts In decentralized systems. Research shows that the responsibility of safeguarding private keys and the unrecoverable key loss reduce the user participation. Specially in general public where inclusivity is essential [2], [3], [4]. The literature further shows that concepts such as self-sovereign identity and identity management remain poorly understood by most users. Studies report unsafe practices and misunderstanding of recovery mechanisms when users are presented with seed phrases or key-based authentication [5], [6]. These findings motivate the need for interface-level abstractions that preserve blockchain security while reducing cognitive load for citizens and officials.

Transaction Latency and User Feedback

Blockchain transactions are inherently asynchronous with confirmation times dependent on network conditions and consensus protocols. This latency can lead to confusion among the users because of the system status problems and users can determine those as system failures [2]. The literature recommends to use an optimistic user interface pattern, where user actions are acknowledged immediately while blockchain confirmation occurs in the background.

As advanced design solutions, literature shows progressive status disclosure, distinguishing between off-chain receipt of a report and on-chain verification which allows users to track the lifecycle of an issue without being exposed to unnecessary low level blockchain details [7], [8]. Mechanisms like that are particularly relevant for issue reporting systems that require transparency and trust.

Transaction Costs and Abstraction of Blockchain Complexity

A significant usability challenge coming from the blockchain is transaction cost that associated with public blockchains. And also the concepts of gas fees and fluctuating gas costs are very unfamiliar in public services context. Most of the time users expect free access to the reporting platforms[1]. The literature strongly suggest that the use of gas fee abstraction techniques. Meta Transactions is one of the example. In Meta, backend relay services submit transactions on behalf of the user. Its achieved by preserving cryptographic intent verification[2]. Therefore this approach enable citizen to interact with blockchain without technical or economical complexities.

Comparison with Centralized Reporting Platforms

FixMyStreet and SeeClickFix are existing centralized platforms. Eventhough they are centralized, they provide a valuable benchmarks for crowdsource citizen reporting interfaces. Intuitive map based reporting, simple authentication mechanisms and clear feedback loops are the key characteristics that attributed to platform success[9], [10].

With addition to that SeeClickFix further uses a gamification mechanism to encourage the users to participate in the system. In contrast, Even though the proposed application uses blockchain for reporting platform, it must retain the usability and the expectations of the users. The literature highlights the need of interface design such that gurantees the data integrity and auditability using familiar ui elements rather than cryptographic representations[11], [8], [12].

Accessibility and Inclusivity Considerations

Inclusivity is a fundamental requirement for e-governance systems. According to [13], it can be seen that poorly designed blockchain interfaces may digitally divide users. Also it may favor crypto-literate users excluding vulnerable populations. To reduce this issue, literature suggests mobile first design approaches. Furthermore, lightweight client architectures, efficient bandwidth synchronization can be used to support access in resource limited environments. [14].

1.4.7 AI-Based Content Moderation in Civic Reporting

Since blockchain is inherently immutable it introduces a unique challenge for content moderation in a civic reporting system. In centralized platforms, administrators can remove or edit inappropriate content. But in a decentralized blockchain based system data cannot be altered once confirmed [15]. While immutability strengthens transparency and accountability, it also raises the risk of permanently storing harmful, illegal, or misleading content. Therefore the literature focuses the need for pre moderation strategies to prevent unsuitable content from being recorded on chain or to limit its visibility after publication [7].

1.4.8 Need for Automated Content Moderation

Civic reporting platforms are susceptible to several forms of misuse that can compromise system reliability and public trust. Automated spam submissions, the submission of malicious or legally sensitive content, and deliberate dissemination of false or misleading reports are some of the common issues [16], [17], [18]. Manual moderation is impractical for above situations. Since the reporting system is large latency, human bias, and operational costs can occur. Furthermore it reintroduces the centralized control structure that blockchain system is aimed to minimize. Therefore the literature supports AI assisted moderation as a scalable and efficient approach for filtering content before its permanent recording. It maintains the acceptable levels of decentralization [19], [20].

1.4.9 Architectural Approaches for AI Blockchain Integration

Due to the computational and storage limitations of blockchain platforms, AI-based moderation cannot be performed directly on chain. Off chain processing combined with on chain verification identified by existing researches as most practical approach [20], [21].

In this approach, reported content is first analyzed by AI models hosted off chain infrastructure or decentralized compute networks. The outcome of this analysis is a validation decision or a trust score. This outcome then submitted to the blockchain as a cryptographic proof or oracle response. Next the smart contracts verify those results before accepting the report. It will ensure that only the content meeting predefined policies are recorded on chain. BlockCITE is a example framework demonstrate how this separation preserves blockchain integrity while avoiding excessive on chain computation [22].

1.4.10 AI Oracles and Decentralized Oracle Networks

Oracles are trusted intermediaries that connect on chain and off chain. In here the AI oracles provides the computational results to smart contracts. Instead of simply retrieving external data to the blockchain, AI oracle perform content classification and verification tasks. Then it returns the structured results to the blockchain [23], [24].

There can be a risk of oracle bias or failure. To minimize the risk a decentralized oracle networks aggregate outputs from multiple independent AI nodes and determine consensus outcomes. This model reduces reliance on a single AI system. Therefore it improves te robustness by requiring agreement among multiple validators[24], [25].

Techniques for Content Analysis

AI-based moderation techniques vary based on the type of submitted content. Natural Language Processing (NLP) models are widely used to detect spam, hate speech and misinformation for text based reports. Studies report high accuracy when combining deep learning models with semantic embeddings. It makes them suitable for filtering large volumes of text submissions prior to blockchain recording [16].

For image based reports computer vision models are used to identify inappropriate contents. However recent advances in generative AI increase the risk of fabricated images. This literature proposes a hybrid verification pipeline that combine cryptographic provenance mechanisms with AI-based image analysis. This ensue that the submitted media is both authentic and contextually valid [26].

Hybrid Human AI Moderation Models

AI enable the scalable moderation. But it may struggle with contextual interpretation and edge cases. Because of that many studies advocate hybrid moderation models. In those models AI performs initial filtering and prioritization, while human reviewers handle disputed or ambiguous cases [19], [27].

Decentralized community based moderation further complement this approach. The models using community annotation systems, allow users or designated validators to challenge AI decisions during predefined review periods. In Addition to that dispute resolution handled by transparent governance mechanisms [7], [28]. Therefore these strategies reduce the reliance on central authority while maintain the accountability.

Ethical Considerations and Content Permanence

Ethical concerns are crucial when implementing systems by integrating AI driven moderation systems. The algorithmic bias and the accountability are the highlights in the literature. Biased training data can lead to unfair rejection of legitimate reports[29]. To minimize this risk explainable AI mechanisms are recommended. This enable users to understand the moderation decision and resubmit corrected reports [30].

Finally, the issue of “toxic immutability” remains a critical challenge for blockchain-based systems. To resolve this issue its suggested that to use a off chain storage systems like IPFS. If the content is later identified as illegal or harmful, it can removed from the off chain storage. But can preserve the immutable audit trail by recording it on the blockchain[26]. The transparency, legal compliance and ethical responsibility can be balanced by this approach.

1.5 Report Outline

Briefly describe what each chapter contains.

Usually 1 page maximum.

Chapter 2

Background

2.1 Blockchain Technology

Blockchain is a distributed, digital, append-only ledger technology. It enables data to be stored in immutable and transparent manner across multiple nodes in a network. Unlike centralized systems, blockchain systems maintain a synchronized copy of the ledger among all authorized participants. This approach reduces the dependency on a single intermediary controlling authority. Each block in the blockchain contains a set of validated transactions, a timestamp, and a cryptographic hash of the previous block. This hash-chaining mechanism ensures that any alteration to previous recorded data becomes immediately detectable.

Immutability is a key characteristic in blockchain technology. It is computationally impractical to alter or delete a transaction after it is confirmed and appended to the blockchain. The immutable property is significantly important in governance-related systems, because integrity and transparency are a must.

Cryptography plays a significant role in blockchain systems. To generate unique digital fingerprints of data, hash functions are used. To verify ownership and authenticity of transactions, digital signatures are used. By using above cryptographic methods, blockchain systems guarantee integrity, authenticity and non-repudiation of data.

2.2 Permissioned Blockchain and PoA

Bitcoin and Ethereum are permissionless blockchains, where anyone can participate as a node and validate transactions. In contrast, permissioned blockchains restrict participation to a predefined set of entities. These entities, often referred to as validator nodes. They are responsible for validating transactions and maintaining the integrity of the blockchain.

While permissionless blockchains rely on consensus mechanisms like Proof of Work or Proof of Stake, the Proof of Authority (PoA) consensus mechanism is specially designed for permissioned blockchains. In here instead of relying on computationally expensive mining or staking process the validator nodes in the permissioned blockchain are validating blocks. Therefore this approach is a reputation and identity-based mechanism, where the authority of the validators is based on their identity and reputation within the network. This allows for faster transaction processing and lower energy consumption compared to traditional consensus mechanisms, making it suitable for enterprise applications and private networks. In the context of local governance perspective, such system requires predictable performance, immutability, transparency and accountability. PoA consensus running in a permissioned blockchain is well suited for these requirements because validators' identities are known and they are controlled by trusted entities. With addition to that, PoA eliminates the need for economic incentives. This makes it more efficient and practical for local governance applications.

2.3 Smart Contracts

Smart contracts are first introduced by Ethereum blockchain. They are self-executing contracts with the terms of the agreement directly written into code. In a blockchain based governance systems smart contract can be used to eliminate centralized administrative control. It can automate workflows that would traditionally require human intervention. Therefore this reduces the reliance on intermediaries, hence improves the transparency and trustworthiness.

The proposed system uses Solidity based smart contracts deployed on a private permissioned PoA blockchain.

2.4 Decentralized Applications (DApps)

Decentralized Applications (DApps) are software applications that operate on a decentralized network, such as a blockchain, rather than being hosted on centralized servers. Unlike traditional web applications that rely on centralized databases and backend infrastructure, DApps utilize smart contracts to handle backend logic and state management. This shift from traditional Web2 to Web3 architecture ensures that application data is immutable, transparent, and resistant to single points of failure or centralized control.

In a typical DApp architecture, the frontend provides the user interface, while the core logic is distributed across the blockchain network. To interact with the blockchain, the frontend utilizes Web3 provider libraries, such as Web3.js or Ethers.js. These libraries enable the application to communicate with blockchain

nodes via Remote Procedure Calls (RPC). Through this connection, the frontend can query the blockchain to read the current state of smart contracts or construct new transactions. When a user wishes to modify the blockchain state, they must cryptographically sign the transaction using their digital wallet's private key, ensuring authenticity and authorization.

A significant challenge in traditional DApp adoption is the requirement for users to pay transaction fees, commonly known as "gas," using the native cryptocurrency of the network. This creates a barrier to entry for regular users who may not possess or understand cryptocurrency. To address this usability issue, modern DApps employ meta-transactions and relayers. A meta-transaction allows a user to cryptographically sign their intent to execute a transaction off-chain without broadcasting it themselves. This signed message is then sent to an intermediary service known as a relayer. The relayer, which holds a balance of the network's cryptocurrency, wraps the user's signed message in a standard blockchain transaction, submits it to the network on behalf of the user, and pays the associated gas fees. This approach provides a seamless, "gasless" user experience while maintaining the security and non-repudiation guarantees of cryptographic signatures, making blockchain applications more accessible to the general public.

2.5 InterPlanetary File System (IPFS)

The InterPlanetary File System (IPFS) is a peer-to-peer network protocol designed for storing and sharing files in a decentralized manner. Unlike the traditional web, where files are identified and accessed through server-based addresses such as URLs, IPFS locates files based on their content. This content-addressed approach improves resilience, reduces dependence on centralized servers, and provides an efficient off-chain storage mechanism for blockchain-based applications. A fundamental concept in IPFS is the Content Identifier (CID), which is a cryptographic hash generated from the content of a file. If the file content changes, the CID also changes, ensuring strong integrity guarantees because users can verify that the retrieved content exactly matches the expected CID. Since identical content generates the same CID, IPFS also supports deduplication and immutable references. In the IPFS network, data can be hosted by any node that chooses to store or pin the content, allowing files to be retrieved from multiple peers using their CID. This decentralized storage model increases availability, improves fault tolerance, and reduces reliance on a single server, although long-term persistence depends on nodes continuing to pin the data. In blockchain-based systems, IPFS is commonly used as an off-chain storage layer because storing large files directly on-chain is costly and inefficient. Instead, applications store only CIDs and essential metadata on the blockchain, while the actual content remains in IPFS. Users can then retrieve

the file through its CID and verify its authenticity using cryptographic hashing. Overall, IPFS provides a scalable and cost-effective decentralized storage solution that combines content integrity, distributed hosting, and efficient off-chain data management.

2.6 Zero-Knowledge and Identity Concepts

Zero-knowledge proofs allow a user to prove that they possess sensitive information without actually revealing the information itself.

The identity decoupling is critical requirement because citizens may hesitate to report sensitive problems they have if their identity is linked to the submitted report. Therefore we decouple identity of the citizen by using public-private key pair associated with the citizen. By leveraging this approach the blockchain ledger will record only pseudonymous identifiers, authorization tickets and cryptographic proofs. This ensure no sensitive personal information is ever ublically exposed.

2.7 AI-based Content Moderation

Citizen reporting platforms are vulnerable to spam Submissions, abusive content, misinformation, and malicious reports. Due to immutability of blockchain records, permanently storing harmful content presents significant ethical and operational concerns. Therefore, content moderation techniques are required before reports are recorded on-chain.

AI techniques are widely used for automated content moderation. Natural Language Processing (NLP) techniques are used in text moderation tasks such as: toxicity detection, hate speech detection, spam filtering, and sentiment Analysis. For image moderation, computer vision and deep learning approaches are mostly used. Convolutional Neural Networks (CNNs) and transformer-based vision models can classify images based on categories like: explicit content, graphic violence, hate symbols, and unrelated media.

Chapter 3

Methodology and Implementation

3.1 Methodology

3.1.1 Research Methodology

This project follows a design-oriented and prototype-based research methodology. The main objective is to design, implement, and evaluate a blockchain-based, community-assisted, privacy-preserving reporting framework for local governance. The system is developed as a functional prototype within a controlled experimental environment rather than as a large-scale production deployment.

The research approach is based on identifying the limitations of existing centralized citizen reporting platforms and designing a decentralized alternative that improves transparency, integrity, anonymity, and trustworthiness. The project combines several technologies, including a private permissioned blockchain, smart contracts, a simulated identity verification mechanism, off-chain storage, AI-assisted content moderation, and a web-based decentralized application.

The system is developed using an incremental and modular development approach. Each major subsystem is designed and implemented independently before being integrated into the complete reporting workflow. This reduces complexity and allows each team member to focus on a specific technical component while maintaining compatibility with the overall architecture.

The major system components are:

1. Web-based DApp
2. ZKP GovID simulator
3. Backend relayer
4. AI moderation oracle service

5. IPFS off-chain storage
6. Smart contracts
7. Private permissioned PoA blockchain

Each subsystem is first implemented and tested individually. After subsystem-level validation, the components are connected through defined APIs, blockchain interfaces, and data formats. The final integration focuses on the end-to-end report submission scenario, where a citizen submits a report through the DApp, the report is validated and moderated, the content is stored off-chain, and the final metadata is recorded on-chain.

This approach follows the phased methodology described in the project proposal, where requirement analysis, system design, blockchain implementation, smart contract development, identity integration, IPFS integration, AI moderation, DApp development, testing, and evaluation are carried out incrementally.

3.1.2 System Requirements Analysis

Functional Requirements

The functional requirements define the main services that the system must provide.

1. Citizen authentication: A citizen should be able to authenticate using the simulated GovID identity service.
2. Ticket issuance: After successful verification, the identity service should issue a citizen seed and a batch of one-time ticket IDs.
3. Report creation: A citizen should be able to create a report containing text, category, location, timestamp, and optional media.
4. Payload signing: The DApp should sign the report payload using the citizen's private key before sending it to the relayer.
5. Relayer verification: The backend relayer should verify the ticket, citizen signature, payload hash, and file constraints.
6. AI moderation: The report content should be analyzed by AI moderation services before blockchain submission.
7. Off-chain storage: Accepted report content and media files should be uploaded to IPFS.

8. Blockchain submission: The relay should submit accepted report metadata to the reporting smart contract.
9. Report retrieval: The DApp should allow citizens and authorized users to read submitted reports and their statuses from the blockchain.
10. Government action workflow: The system should support future integration of government-side actions such as issue review, resolution updates, and closure.

Non-functional Requirements

The system also has several non-functional requirements.

1. Security: The system should use hashing, digital signatures, ticket validation, role-based access control, and secure API communication.
2. Privacy: Citizen identities should be decoupled from on-chain report records.
3. Integrity: Report payloads, IPFS content, oracle decisions, and blockchain records should be tamper-evident.
4. Availability: The deployed services should remain accessible during testing and demonstration.
5. Performance: The system should process reports within reasonable time under prototype workloads.
6. Maintainability: The system should be modular and containerized to simplify deployment and debugging.
7. Auditability: Important workflow decisions should be logged and anchored through hashes or blockchain events.

3.1.3 Proposed System Architecture

Overall System Architecture

The proposed system uses a hybrid decentralized architecture that combines on-chain and off-chain components. The blockchain is responsible for storing immutable metadata, workflow states, and references to report content. Computationally intensive and large-data operations are handled off-chain through the relay, AI moderation service, and IPFS. The overall architecture of the system is shown in the Figure [3.1](#)

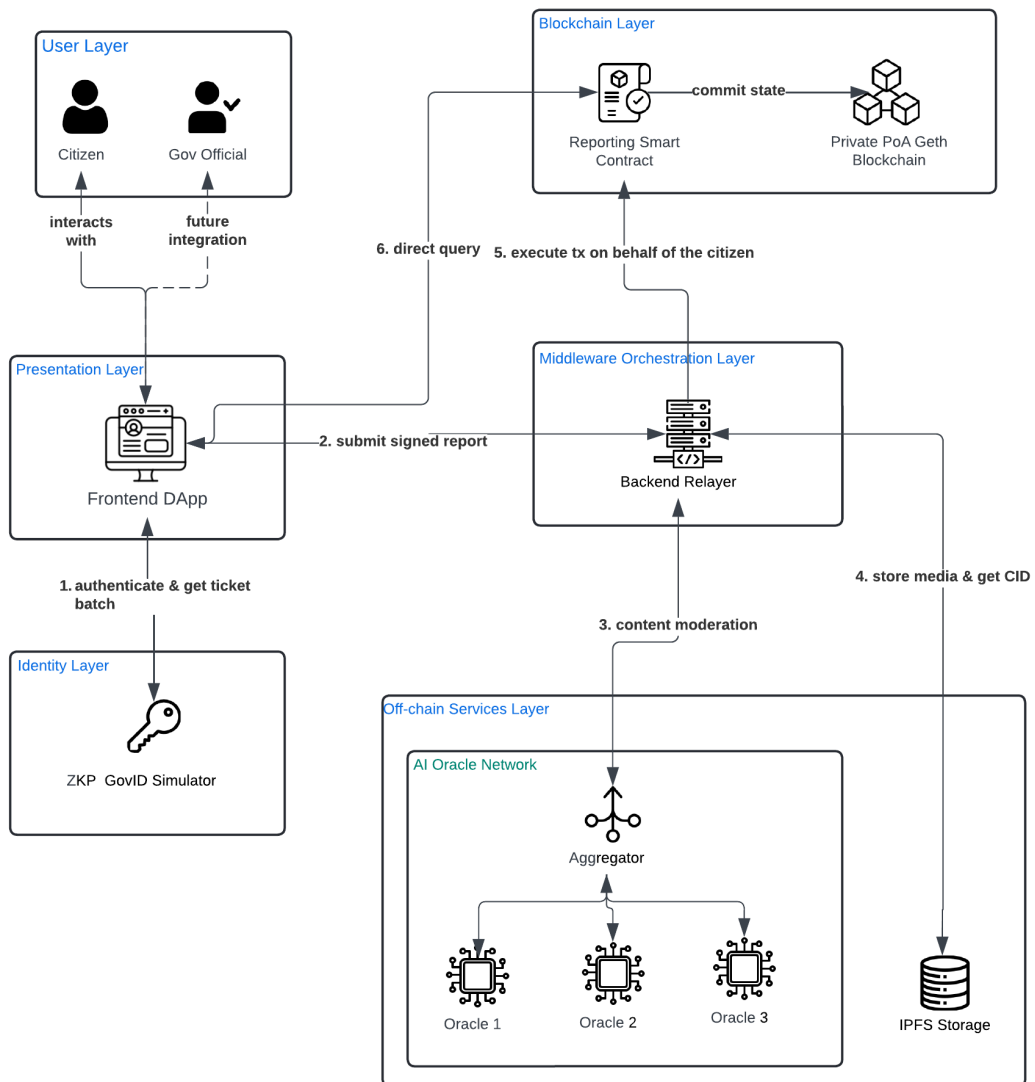


Figure 3.1: High Level Architecture Diagram.

The DApp acts as the citizen-facing interface. The ZKP GovID simulator handles identity verification and ticket issuance. The backend relay verifies signed report submissions and coordinates communication between AI moderation, IPFS, and smart contracts. The AI moderation service evaluates report content before storage and blockchain submission. IPFS stores large report data and media evidence, while the smart contracts manage the report lifecycle on the private PoA blockchain.

This architecture supports the main project goal of preserving citizen anonymity while maintaining report integrity and transparency. The use of a relay also improves usability because citizens do not need to directly manage gas payments or manually submit blockchain transactions.

Layered Architecture

The system can be divided into six major layers.

1. User Layer: This layer consists of the citizens and government officials.
2. Presentation layer: This layer consists of the DApp used by citizens and government officials. It provides interfaces for login, report submission, report viewing, and future government-side actions.
3. Identity and authentication layer: This layer consists of the ZKP GovID simulator. It verifies citizens and issues citizen seeds and ticket batches. This separates real identity verification from on-chain activity.
4. Middleware orchestration layer: The backend relay acts as the coordination layer. It receives signed report payloads, performs validations, calls AI moderation services, uploads accepted content to IPFS, and submits blockchain transactions.
5. Off-chain services layer: This layer includes the AI moderation service and IPFS storage. These components handle tasks that are unsuitable for direct blockchain execution.
6. Blockchain layer: This layer consists of the smart contracts and the private PoA blockchain. It stores immutable metadata, report states, content hashes, IPFS CIDs, and workflow events.

3.1.4 Proposed Workflow

Citizen Authentication and Identity Workflow

Report Submission Workflow

3.1.5 Project Plan and Timeline

3.2 Project Timeline

The project is planned to be completed over two academic semesters following a structured and incremental development approach as shown in Figure 3.2. The timeline allocates sufficient time for system design, implementation, testing, evaluation to ensure the successful completion of the proposed objectives.

3.2.1 Expected Outcomes

The expected outcome of the project is a working prototype of a privacy-preserving, blockchain-based citizen reporting system for local governance.

The system is expected to achieve the following outcomes:

1. Anonymous civic reporting: Citizens can submit reports without directly revealing their real-world identity on-chain.
2. Tamper-evident records: Report metadata, hashes, and workflow states are recorded on a private blockchain.
3. Improved report reliability: AI-assisted moderation helps detect spam, abuse, and irrelevant submissions before blockchain anchoring.
4. Scalable storage: Large report content and media evidence are stored in IPFS, while blockchain stores only metadata and references.
5. Usable blockchain interaction: The relayer pattern removes the need for citizens to directly manage gas fees or blockchain transactions.
6. Modular deployment: The system is designed as a collection of independently deployable components.
7. Governance transparency: The blockchain provides an auditable record of report submission and status changes.

Through these outcomes, the prototype demonstrates the feasibility of combining blockchain, privacy-preserving identity mechanisms, off-chain storage, AI-assisted moderation, and web-based DApp interfaces to support transparent and accountable local governance reporting.

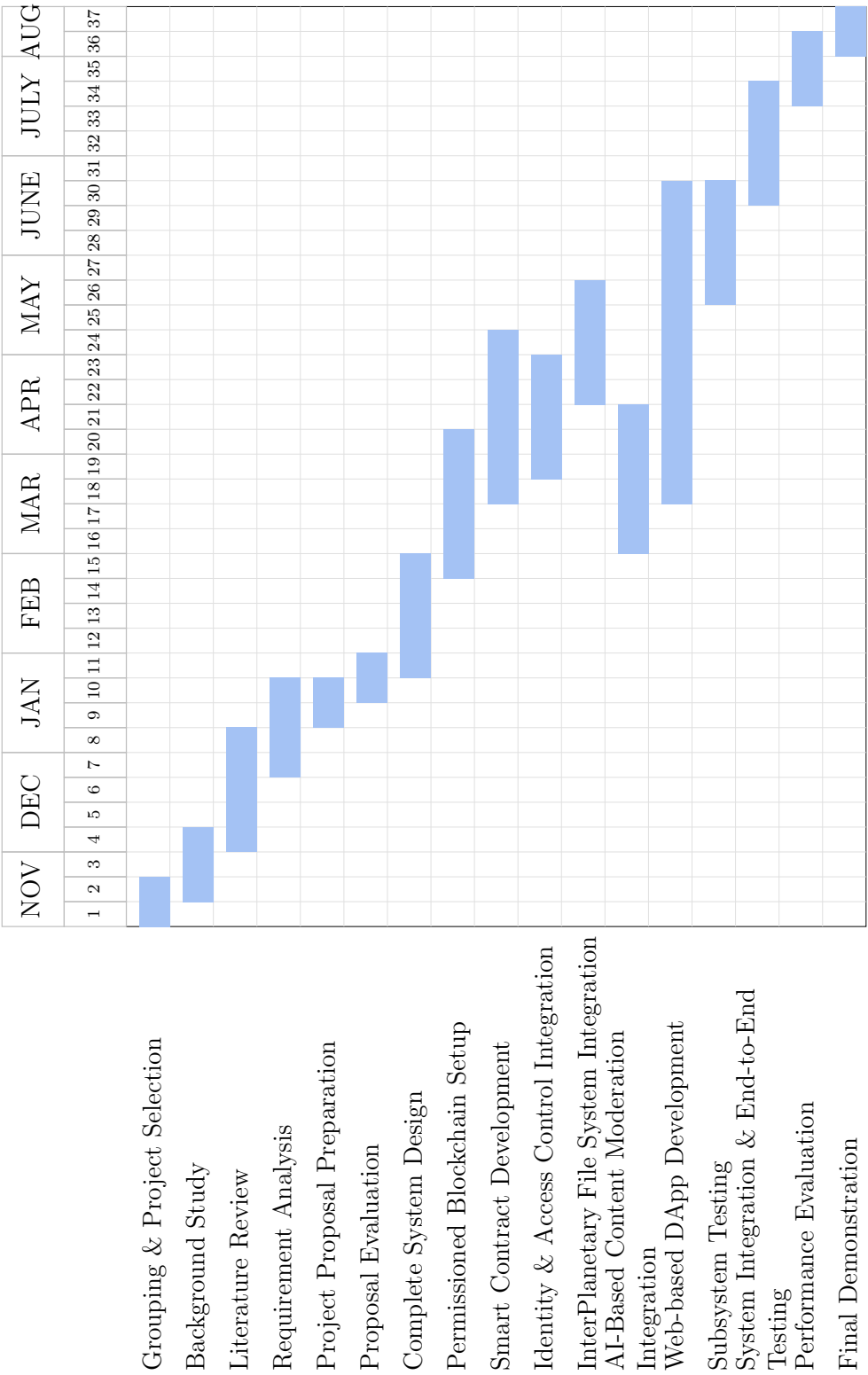


Figure 3.2: Project Timeline Gantt Chart

3.3 Implementation

3.3.1 Permissioned Blockchain Implementation

The core infrastructure of the proposed system is the private permissioned blockchain network. The implementation focused on setting up a secure and efficient environment tailored for local governance applications.

- **Geth PoA Network Setup:** A private permissioned Ethereum-compatible network was initialized using the Go Ethereum client. The network was configured to operate in proof-of-authority consensus mode, specifically using the Clique algorithm. This enabled faster block confirmation and lower energy consumption compared to traditional proof-of-work systems. A custom genesis file was created to define network parameters, including the initial set of authorized signers and the chain ID.
- **Validator Node Configuration:** Minimum of three full nodes were required to act as an validator. The 3 nodes are deployed in a personal vps infrastructure. Each node is configured with a unique cryptographic identity and their addresses were added to the list of authorized sealers within the consensus engine to manage transaction validation authority.
- **Network Security Measures:** To communicate with the blockchain network, secure Remote Procedure Call (RPC) endpoint was established only for node1. Additionally, the network is configured to operate in a private subnet, further isolating it from external threats. This configuration allowed DApp and smart contracts to communicate with ledger enabling functions such as report submission, transaction querying and state synchronization.

3.3.2 Smart Contract Architecture and Logic

The core on-chain logic of the decentralized reporting framework is governed by the Reporting.sol smart contract. This contract is designed to be highly gas-efficient, acting primarily as a state machine and a cryptographic ledger rather than a data storage layer. The implementation is divided into three critical functional domains as The Report Contract structure, Ticket Validation, and State Transition Logic.

The Report Contract

The Report Contract is engineered using a Separation of Concerns principle. To mitigate the prohibitive gas costs associated with storing large multimedia files on the blockchain, the contract only stores essential metadata.

1. **Data Structure:** Each civic issue is stored in a Report struct containing a unique integer ID, the IPFS Content Identifier (CID) pointing to off-chain media, voting counters, the assigned authority address, and its current lifecycle state.
2. **Role-Based Access Control (RBAC):** Utilizing OpenZeppelin's AccessControl, the contract explicitly defines operational boundaries. The RELAYER_ROLE is restricted to the backend server, allowing it to submit reports and relay votes on behalf of citizens. The AUTHORITY_ROLE is granted to recognized government or NGO nodes, granting them permission to update an issue's status to solved or rejected.

Ticket Validation and Sybil Resistance

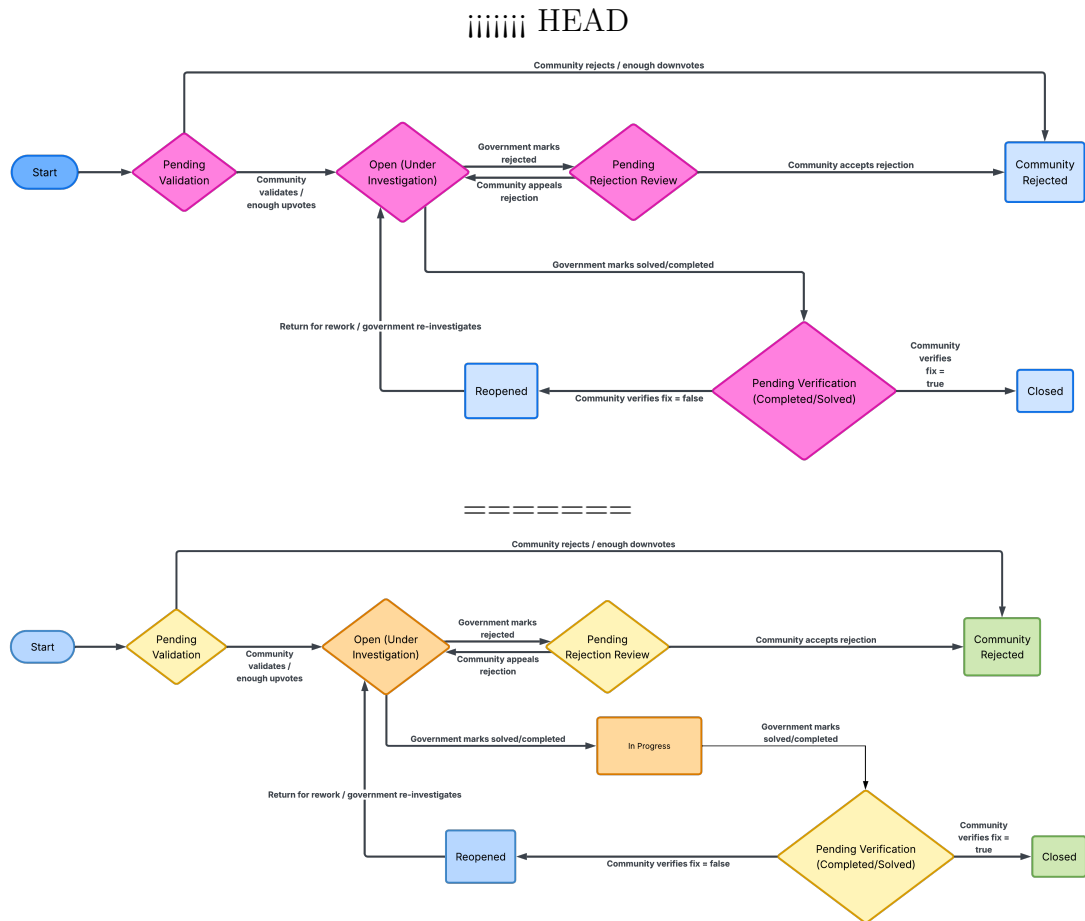
A fundamental challenge in anonymous decentralized systems is preventing Sybil attacks and duplicate submissions. This framework utilizes a dual-layer validation mechanism based on Zero-Knowledge Proof (ZKP) tickets.

1. **Off-Chain Cryptographic Validation:** Before a transaction reaches the blockchain, the backend relayer intercepts the payload. Using ethers.js, it verifies that the zkpTicketId possesses a valid cryptographic signature from the official Government Authority. It subsequently verifies the citizen's signature against the payload hash to ensure data integrity during transmission.
2. **On-Chain Nullifier Tracking:** Once validated off-chain, the zkpTicketId is passed to the smart contract as a submissionNullifier. The contract maintains a mapping of public submissionNullifiers. Upon successful report creation, this nullifier is marked as used. Any subsequent attempt to submit a report using the same ticket will systematically fail, cryptographically guaranteeing that one authorized ticket equates to exactly one unique report.

State Transition Logic (Finite State Machine)

To enforce accountability and prevent unilateral dismissals of civic issues by authorities, the contract implements a strict Finite State Machine (FSM). Every submitted report strictly adheres to an 8-stage lifecycle managed via an on-chain enum:

1. **Pending Validation:** Initial state awaiting community confirmation of legitimacy.
2. **Community Rejected:** Terminal state for spam or invalid issues.
3. **Open:** Validated by citizens and actively awaiting government action.



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Figure 3.3: State transition diagram for the civic reporting system.

4. In Progress: The authority has officially acknowledged the issue and is actively executing physical repairs or resolution steps.
5. Pending Rejection Review: The authority dismissed the issue, triggering a community appeal phase.
6. Pending Verification: The authority marked the active work as completed, triggering a mandatory community verification vote.
7. Closed: Terminal state indicating a successfully verified resolution or an accepted rejection.
8. Reopened: The community rejected the authority's fix, forcing the issue back to the authority for rework.

A report must traverse these specific states driven by a combination of authority actions and community consensus thresholds.

1. Community-Driven Transitions: When a report is submitted, it enters the Pending Validation state. The transition to the Open state (making it visible for government action) only occurs automatically via the `_changeStatus()` internal function once community upvotes reach the predefined `VALIDATION_THRESHOLD`. Conversely, reaching the `REJECTION_THRESHOLD` moves it to the terminal Community Rejected state.
2. Authority-Driven Transitions: Authorities can only interact with reports in the Open, Reopened, or In Progress states. An authority calling `startWork()` shifts an Open or Reopened issue to the In Progress state, providing transparency to citizens that action is underway. From the In Progress state, calling `markAsSolved()` shifts the report to Pending Verification. Additionally, calling `rejectIssue()` from the Open or In Progress states shifts it to Pending Rejection Review.
3. Consensus Verification: The FSM mandates that authority actions are never final until verified by the community. If an authority marks an issue as solved, the community must vote to verify the physical fix. Achieving the verification threshold transitions the report to the terminal Closed state. If the community rejects the fix, the state transitions to Reopened, forcing the authority to address the issue again.

3.3.3 Identity and Access Control Implementation

To satisfy the project's requirement for privacy-preserving accountability, an identity decoupling mechanism was successfully developed. This subsystem ensures

that citizens can interact with the blockchain anonymously while remaining verified by an off-chain authority. The implementation was structured into three main components

- **GovID Simulator:** An off-chain simulated identity server was developed and deployed to a personal Virtual Private Server (VPS). This service acts as the trusted authority for verifying real-world identities. A dedicated API endpoint (`/api/govid/verify-citizen`) was implemented to securely authenticate users via their National Identity Number (`govId`) and a password, bridging the gap between traditional web authentication and decentralized access.
- **Citizen Keypair generation:** To completely decouple a user's real identity from their on-chain footprint, the identity server does not store or transmit blockchain addresses. Instead, upon successful authentication, the simulator generates and returns a secure, 256-bit `citizenSeed`. This cryptographic seed is then used by the client-side DApp to deterministically generate an anonymous public-private keypair, allowing the citizen to interact securely with smart contracts without exposing personal data.
- **Ticket Batch Generation:** To facilitate multiple secure, anonymous report submissions over time, the simulator implements a batch authorization protocol. During authentication, the server generates a structured `ticketBatch` (currently configured to issue 10 tickets per request). Each ticket contains a unique `ticketId` (a cryptographic hash) and a signature signed by the Identity Authority's private key. Citizens use these single-use tickets to authorize their transactions on-chain. The backend relayer validates the authority's signature on the ticket, granting access without ever knowing which specific citizen is submitting the report.

3.3.4 Backend Relayer Implementation

The backend relayer serves as a critical middleware component that bridges the frontend DApp with the blockchain network and external services. It is responsible for handling report submissions, managing interactions with IPFS for off-chain storage, and orchestrating AI-based content moderation before anchoring data on the blockchain.

The implementation utilizes the NestJS framework to provide a robust and highly modular RESTful API. The core workflow of the relayer executes the following deterministic pipeline for each incoming citizen report.

1. **Cryptographic Verification:** The relayer receives the payload via a `POST /report` endpoint. Utilizing the `ethers.js` library, it performs a strict dual-signature validation:

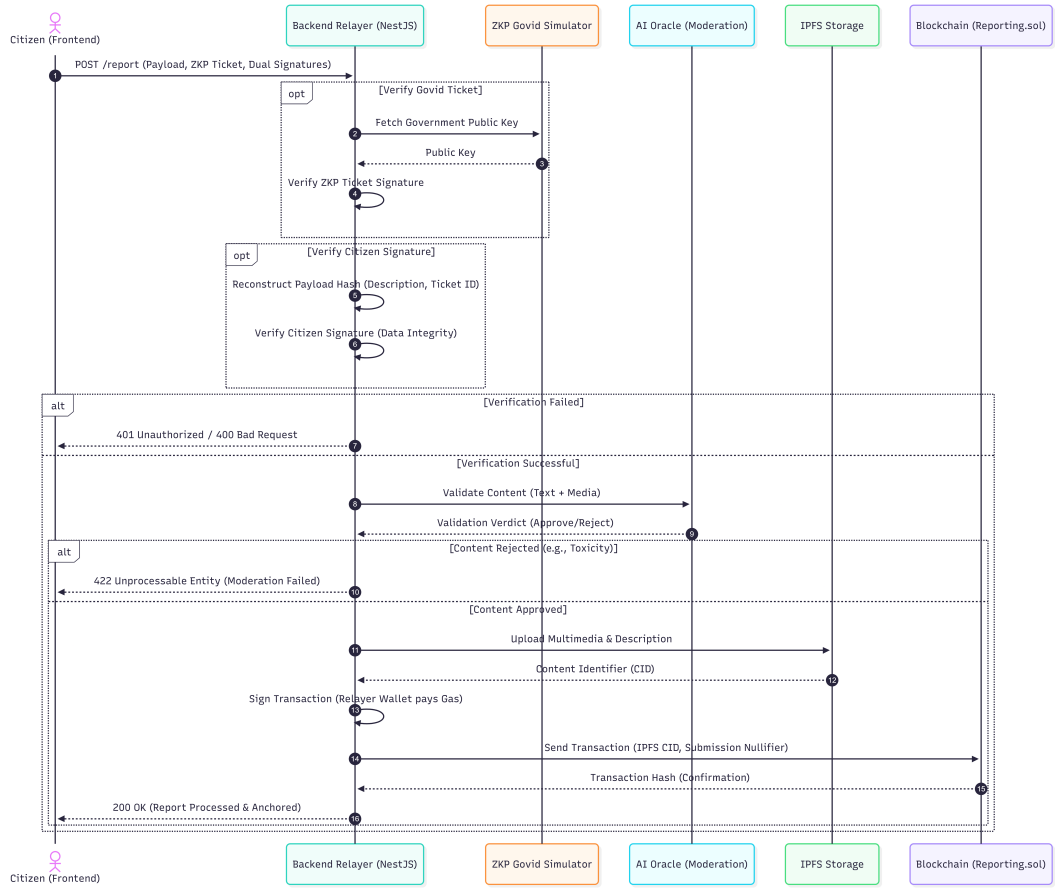


Figure 3.4: Backend relayer workflow architecture.

- (a) It verifies the ZKP Govid Simulator ticket signature against the dynamically fetched Government Authority public key to ensure the user is an authorized citizen.
 - (b) It reconstructs the payload hash (binding the description and ticket ID) and verifies the citizen's signature to prevent data tampering in transit.
- 2. **AI Content Moderation:** To prevent on-chain spam, toxic immutability of blockchains and ensure platform integrity, the relayer routes the submitted text and media to the AI Oracle moderation service. This service acts as an automated filter, analyzing the content for appropriateness and returning a validation verdict before any storage operations occur.
- 3. **Decentralized Storage (IPFS):** To maintain a highly efficient and cost-effective blockchain state, heavy multimedia files and detailed text are offloaded to the InterPlanetary File System (IPFS). The relayer manages the file buffers via multer, processes the upload, and retrieves an immutable Content Identifier (CID).
- 4. **Gasless Blockchain Submission:** As the final step, the relayer formats a transaction containing the IPFS CID and the ZKP ticket (which serves as a Sybil-resistant submissionNullifier). The relayer signs and dispatches this transaction to the Reporting.sol smart contract. By paying the network gas fees from its own wallet, the relayer enables a completely frictionless, gasless experience for the end-user.

3.3.5 IPFS Off-chain Storage Integration

The implementation of IPFS off-chain storage in the proposed system enables secure, decentralized, and efficient management of files through the use of Content Identifiers (CIDs) and distributed storage mechanisms. When a user uploads a file through the application interface, the backend service communicates with the IPFS network and adds the file to IPFS. During this process, IPFS automatically generates a unique CID based on the cryptographic hash of the file content. This CID serves as a permanent and tamper-proof reference to the uploaded file because any modification to the content results in a completely different CID. Instead of storing the entire file directly on the blockchain, only the CID and related metadata, such as file type, uploader information, and timestamps, are stored in the application database or on-chain record. This approach significantly reduces blockchain storage costs and improves scalability while still allowing files to be verified and retrieved through their associated CIDs. The actual file remains stored in a decentralized manner across the IPFS network, where multiple nodes may pin and host the content to ensure availability and fault tolerance. When a file is requested, the

system retrieves it from IPFS using the CID, and the network locates the content among participating peers and delivers it while preserving integrity. Within the project architecture, the integration is managed through backend services responsible for communication with the IPFS node, while controller components handle API requests related to file uploads and retrieval operations. Additional middleware components are incorporated for file validation, request processing, and error handling to ensure reliable system functionality. The overall workflow involves uploading a file to IPFS, generating and storing the CID with metadata, and later retrieving the file from the network using the CID whenever access is required. Through this implementation, IPFS provides a reliable off-chain storage solution that combines decentralized hosting, strong integrity verification, reduced on-chain storage requirements, and efficient file management.

3.3.6 AI-based Moderation System Implementation

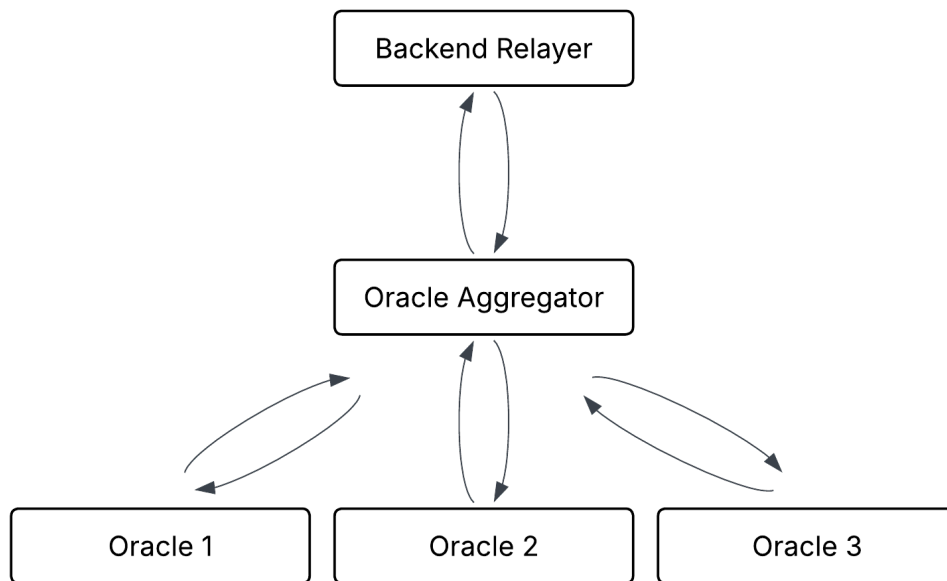


Figure 3.5: Oracle network architecture for AI moderation.

The AI moderation subsystem is designed as an off-chain oracle network. The moderation process is performed off-chain and the result is returned to the reporting workflow. The oracle network consists of three independent oracle workers and one aggregator. The aggregator forwards the moderation request to all oracle

workers, collects their decisions, applies a voting rule, signs the final decision, and returns the result to the relayer. This architecture is demonstrated in Figure 3.5. The oracle 1 focuses on detecting toxic, abusive, threatening, or harmful language. It analyzes the submitted report using a predefined moderation strategy and returns a vote. The oracle 2 focuses on spam detection. It checks for promotional content, repeated characters, and extremely short descriptions. The third oracle focuses on civic relevance. It checks whether the report is related to local governance issues such as roads, garbage, water, drainage, public infrastructure, and community facilities.

The oracle aggregator combines the decisions of the three oracle workers. Each oracle returns either ACCEPT or REJECT. The current aggregation logic uses a two out of three (2/3) majority rule. The final moderation result includes the final decision, confidence score, report hash, oracle votes, decision hash, and aggregator signature.

To ensure integrity of the moderation results, several mechanisms are used. The aggregator computes a hash of the submitted report payload. The final moderation result and oracle votes are combined and hashed to create the decision hash. Also, the aggregator signs the decision hash using its private key.

3.3.7 Web-based DApp Development

The decentralized application (DApp) serves as the primary gateway for citizens to interact with the local governance system. It is built using a modern web stack comprising Next.js and React for the frontend interface, Tailwind CSS for responsive styling, and Ethers.js for cryptographic operations. The architecture is designed to abstract blockchain complexities away from the end-user while maintaining strict privacy and security guarantees.

User Authentication Interface

The authentication interface is designed to verify citizen eligibility without compromising personal privacy.

- **GovID Verification Integration:** The interface securely collects the citizen's GovID credentials and interfaces with an external GovID Simulator API. This abstracts the process of verifying a citizen's real-world identity against a centralized registry.
- **Cryptographic Credential Generation:** Upon successful identity verification, the system retrieves a batch of Zero-Knowledge Proof (ZKP) tickets and a unique 'citizenSeed'.

- **Local State Management:** Using the ‘ethers.js’ library, a localized Citizen Wallet is derived from the seed. A custom React Context (`CitizenContext`) is utilized to manage the session state, ensuring that the generated private keys and ZKP tickets remain strictly on the client-side device and are never exposed to external servers. The UI incorporates dynamic feedback mechanisms to indicate the generation of cryptographic proofs to the user.

Report Submission Interface

The report submission module allows authenticated citizens to report civic issues anonymously.

- **Form Design and Data Handling:** Developed with a mobile-first approach, the form captures issue descriptions and optional photographic evidence using standard `FormData` structures to support multipart uploads.
- **Anonymous Cryptographic Signing:** To guarantee anonymity and prevent spam, each submission consumes a single ZKP ticket from the user’s allocated batch. The payload (consisting of the issue description and the ZKP ticket ID) is hashed using Keccak-256 and subsequently signed using the citizen’s locally stored ECDSA private key.
- **Meta-Transaction Relaying:** Instead of submitting transactions directly to the blockchain and incurring gas fees, the frontend dispatches the signed payload, image, and ZKP metadata to a NestJS Backend Relayer. The relayer handles the intermediate steps: uploading the evidence to IPFS, requesting AI moderation, and finally submitting the transaction to the Geth blockchain network on behalf of the user.

Blockchain Query Interface

The query interface acts as a community feed where citizens can view, filter, and track the status of reported issues.

- **Mobile-First Feed Architecture:** Implemented as a single-column, scrollable interface inspired by modern social media platforms (e.g., Reddit). This design choice prioritizes readability and user engagement on mobile devices.
- **Dynamic Issue Cards:** Data retrieved from the blockchain is formatted into interactive cards. Each card displays critical metadata including categorical status badges (e.g., PENDING VALIDATION, OPEN, RESOLVED), community upvotes, truncated descriptions, timestamps, and image thumbnails.

- **State Management and Navigation:** The interface employs React state hooks to allow real-time filtering of issues by category (e.g., Infrastructure, Parks & Rec, Safety). Furthermore, a persistent Floating Action Button (FAB) is implemented to provide users with seamless access to the report submission workflow at any time.

Chapter 4

Testing, Experimentation, and Validation

This chapter outlines the proposed methodology for the testing, experimentation, and validation of the decentralized reporting framework. Given the distributed nature of the system—which integrates off-chain cryptographic services, AI moderation, and on-chain smart contracts—a rigorous experimental setup will be established to evaluate performance, security, and functional correctness.

4.1 Experimental Setup

To accurately simulate a real-world deployment of the permissioned blockchain and its ancillary microservices, the experimental environment has been hosted on a cloud-based infrastructure utilizing containerization technologies.

4.1.1 Hardware Infrastructure

The testing environment was deployed across two identical Virtual Private Server (VPS) instances to simulate a distributed network architecture and evaluate inter-service communication latency. The hardware specifications for each VPS are as follows:

1. Compute: 4 vCPU Cores
2. Memory: 8 GB RAM
3. Storage: 75 GB SSD
4. Network: High-bandwidth cloud uplink

Server 1 acts as the primary operational node, hosting the core citizen-facing applications, middleware, and the blockchain ledger. Server 2 is strictly dedicated to hosting the computationally intensive AI moderation ensemble and the IPFS storage layer. This isolation prevents resource monopolization and ensures accurate performance profiling of the blockchain consensus mechanism.

4.1.2 Software and Deployment Environment

To ensure reproducibility and consistency across test iterations, the entire software stack was containerized using Docker.

1. **Orchestration and CI/CD:** Deployment and lifecycle management of the containers are handled by Dokploy, a self-hosted Platform-as-a-Service (PaaS) solution installed on both servers.
2. **Automated Deployment:** Continuous Integration and Continuous Deployment (CI/CD) pipelines were established via GitHub webhooks. Dokploy was configured to monitor the develop branch of the project repository. Upon receiving a trigger, it automatically pulls the latest source code, builds the respective Dockerfiles for the microservices, and deploys the updated containers.

4.1.3 Blockchain Configuration (Server 1)

The on-chain environment consists of a private, permissioned Proof-of-Authority (PoA) blockchain.

1. **Client Software:** The network was initialized using Go-Ethereum (Geth) version 1.13.15.
2. **Consensus Engine:** The Clique PoA consensus algorithm was utilized to ensure deterministic block times and eliminate the computational overhead of Proof-of-Work, which closely aligns with the operational model of a government-backed consortium chain.
3. **Node Topology:** A 3-node cluster was deployed via a unified docker-compose.yml configuration. These nodes serve as the designated Authority (Sealer) nodes representing various governance entities.

4.1.4 Application and Services Topology

The testing setup evaluates the interaction between the following primary subsystems distributed across the two servers.

Deployed on Server 1 (Core Operations):

1. ZKP-GovID Simulator: A Node.js service simulating the issuance of government-backed Zero-Knowledge Proof tickets.
2. Frontend DApp: A Next.js application providing the citizen interface.
3. Backend Relayer: A NestJS-based middleware component responsible for off-chain cryptographic validation, routing, and gasless blockchain transaction submission.
4. Smart Contracts: Compiled using the Hardhat development environment utilizing Solidity version 0.8.20.

Deployed on Server 2 (Storage and AI Oracle):

1. AI Moderation Ensemble: A multi-node AI oracle architecture designed for robust spam and content filtering. This consists of three distinct AI algorithm nodes running concurrently to analyze incoming media and text. These nodes are governed by a central Aggregator Node that acts as the single entry point and determines the final moderation verdict based on algorithmic consensus.
2. IPFS Node: A dedicated InterPlanetary File System (IPFS) node that will be deployed in the subsequent testing phase to handle the distributed, off-chain storage of all civic report multimedia (images) and heavy metadata, returning immutable Content Identifiers (CIDs) to the backend relayer.

4.2 Testing Methodology

To ensure the reliability, security, and performance of the decentralized reporting framework, a comprehensive, multi-tiered testing methodology will be employed. This approach systematically validates the architecture from isolated components up to the complete end-to-end user workflow.

4.2.1 Subsystem Testing

Subsystem testing will focus on isolating individual architectural components to verify their internal logic, cryptographic integrity, and algorithmic correctness prior to network integration.

1. Smart Contract Validation: The core reporting smart contract will undergo rigorous unit testing using the Hardhat development framework. Automated

tests will be written to validate the logic of the Finite State Machine, ensuring that community voting thresholds trigger correct status changes and that role-based access controls prevent unauthorized authority actions.

2. Backend Relayer Verification: The middleware will be tested using a dedicated testing suite. The primary focus will be validating the cryptographic verification algorithms, ensuring that both the Zero-Knowledge Proof tickets from the simulator and the citizen signatures are correctly authenticated or systematically rejected when malformed.
3. AI Oracle Evaluation: The AI moderation ensemble will be tested in isolation using a curated dataset of synthetic civic reports. This dataset will include a mixture of valid public issues, explicit spam, and manipulated imagery to evaluate the accuracy, false-positive rate, and consensus latency of the multi-node aggregator.

4.2.2 Integration Testing

Once individual subsystems are validated, integration testing will be conducted to evaluate the communication interfaces, network latency, and data flow between the distributed microservices across the two virtual private servers.

1. Relayer and Off-Chain Services: Tests will verify the backend relayer's ability to seamlessly orchestrate the initial validation pipeline. This includes measuring the successful routing of payloads to the AI Oracle ensemble, the handling of timeout exceptions, and the successful uploading of multimedia to the IPFS node to retrieve immutable Content Identifiers.
2. Relayer and Blockchain Interface: Integration scripts will simulate the relayer dispatching transactions to the private Proof-of-Authority blockchain network. This phase will verify the gasless submission mechanics, ensuring the relayer accurately wraps the Content Identifiers and ticket nullifiers into valid transactions that are accepted by the network nodes.
3. Frontend to Middleware Communication: API integration tests will confirm that the decentralized application securely formats and transmits the cryptographic payloads, including the user description and attached media, to the relayer over encrypted channels.

4.2.3 Functional Testing

Functional testing will evaluate the entire integrated system against the defined operational requirements from an end-user perspective, simulating both real-world

usage and adversarial conditions.

1. **End-to-End User Journeys:** Automated scripts will simulate complete, valid report lifecycles. This entails a simulated citizen generating a ticket, submitting a valid issue via the frontend, passing AI moderation, anchoring on the blockchain, and successfully progressing through the community validation and authority resolution states.
2. **Sybil Resistance and Replay Attack Mitigation:** The framework will be subjected to adversarial functional tests. Scripts will intentionally attempt to submit multiple reports using identical ticket nullifiers to verify that the smart contract successfully rejects duplicate entries and maintains the one-ticket-one-vote invariant.
3. **Cryptographic Forgery Attempts:** The system will be tested by injecting payloads with manipulated descriptions or mismatched signatures. This will ensure the backend relayer systematically drops tampered requests before they can interact with the off-chain storage or consume blockchain resources.

4.3 Blockchain Performance Testing

Metrics:

latency, throughput, block confirmation.

4.4 Smart Contract Validation

Test:

report creation, state transitions, ticket validation.

4.5 AI Moderation Testing

AI moderation testing evaluates whether the moderation subsystem correctly classifies submitted report content as acceptable or unacceptable. At the current stage, the implemented AI moderation service focuses mainly on text-based moderation. Image moderation is planned as a future extension.

The objectives of AI moderation testing are:

- to verify that the oracle aggregator API is accessible,
- to verify that all three oracle workers respond correctly,

- to test valid civic report acceptance,
- to test spam and abusive content rejection,
- to verify majority voting behavior,
- to verify generation of report hash, decision hash, and signature.

The AI moderation service is deployed on an Ubuntu VPS 2 using Dokploy and Docker containers. The deployment includes:

- Government Oracle container,
- International Oracle container,
- NGO Oracle container,
- Oracle Aggregator container.

The backend relayer communicates only with the aggregator API. The aggregator communicates with the three oracle workers using internal container networking.

4.5.1 API Availability Test

The test cases used for testing moderation API availability are shown in the Table [4.1](#)

Table 4.1: AI moderation API availability tests

Test ID	Test	Expected Result	Status
AI-T01	Call aggregator health endpoint	Service returns running status	Passed
AI-T02	Check oracle worker availability	Aggregator identifies available oracle workers	Passed
AI-T03	Send request without API key	Request rejected	Passed
AI-T04	Send request with valid API key	Request processed successfully	Passed

4.5.2 Moderation Classification Tests

The test cases used for testing moderation classification are shown in Table [4.2](#)

Table 4.2: AI moderation classification tests

Test ID	Input Type	Example Input	Expected Decision
AI-T05	Valid civic report	“There is a large pothole near the public school.”	ACCEPT
AI-T06	Spam content	“BUY NOW FREE MONEY CLICK HERE.”	REJECT
AI-T07	Toxic content	Text containing abusive or threatening phrases	REJECT / REVIEW
AI-T08	Ambiguous report	“Bad situation near town.”	REVIEW
AI-T09	Category mismatch	Garbage issue submitted under road damage category	REVIEW
AI-T10	Empty report	Empty text field	Reject request / validation error

4.5.3 Oracle Voting Validation

The oracle moderation system uses a three-node voting mechanism. Each oracle returns either one of the votes, **ACCEPT** or **REJECT**.

The aggregator applies majority-based decision logic.

Table 4.3: Oracle voting validation

Oracle Votes	Expected Final Decision
ACCEPT, ACCEPT, ACCEPT	ACCEPT
ACCEPT, ACCEPT, REJECT	ACCEPT
REJECT, REJECT, ACCEPT	REJECT
REJECT, REJECT, REJECT	REJECT

4.5.4 Response Structure Validation

Each successful oracle response should include the fields shown in Table 4.4.

The presence of `report_hash`, `decision_hash`, and `aggregator_signature` supports auditability and tamper-evidence in the moderation process.

Table 4.4: Oracle aggregator response fields

Field	Purpose
<code>final_decision</code>	Final moderation result
<code>final_confidence</code>	Aggregated confidence value
<code>report_hash</code>	Hash of submitted report payload
<code>decision_hash</code>	Hash of final moderation decision
<code>oracle_votes</code>	Individual oracle outputs
<code>aggregator_signature</code>	Signature of decision hash

The AI content moderation service was successfully deployed as separate containers on a VPS. Preliminary API testing confirms that the aggregator can receive moderation requests, communicate with the oracle workers, collect votes, and return an aggregated decision. This validates the feasibility of the decentralized oracle moderation approach at prototype level.

4.6 IPFS Storage Testing

To evaluate the effectiveness and reliability of the IPFS off-chain storage integration, several testing and validation procedures were conducted focusing on upload performance, retrieval integrity, and CID validation. The upload time was measured for files of different sizes in order to assess the responsiveness of the system and the overhead associated with decentralized storage. During each test, the start and end times of the upload process were recorded, and the total duration was calculated. The results indicated that smaller files were uploaded quickly, while larger files required proportionally more time due to the distributed nature of the IPFS network. Retrieval integrity tests were performed to confirm that files downloaded from IPFS remained identical to the originals. After uploading a file and obtaining its Content Identifier (CID), the same file was retrieved from IPFS using the CID and compared byte-for-byte with the original file. The successful matches demonstrated that IPFS reliably preserves file integrity during both storage and retrieval operations. In addition, CID validation tests were conducted to ensure that the generated CID accurately represented the corresponding file content. For each uploaded file, the CID produced by IPFS was recorded, and after retrieval, the CID was recalculated from the downloaded file and compared against the original value. Matching CIDs confirmed that the content had not been modified and verified the integrity guarantees provided by the IPFS content-addressing mechanism. Overall, these tests demonstrate that the implemented IPFS integration satisfies key requirements for decentralized storage, including efficient file uploads, reliable retrieval, and strong verification through cryptographic content identifiers.

4.7 End-to-End Integration Testing

To ensure the seamless operation of the entire platform, end-to-end integration testing was conducted by simulating the complete lifecycle of a civic issue report. This testing phase verifies the interoperability between the frontend DApp, the backend relayer, the IPFS storage network, the AI moderation oracle, and the underlying permissioned blockchain.

The end-to-end testing workflow was executed as follows:

1. **Citizen Authentication and Key Generation:** The test commenced with a simulated citizen authenticating via the external GovID simulator. Upon successful verification, the system successfully generated a batch of Zero-Knowledge Proof (ZKP) tickets and derived the local ECDSA wallet, demonstrating the proper functioning of the identity decoupling mechanism.

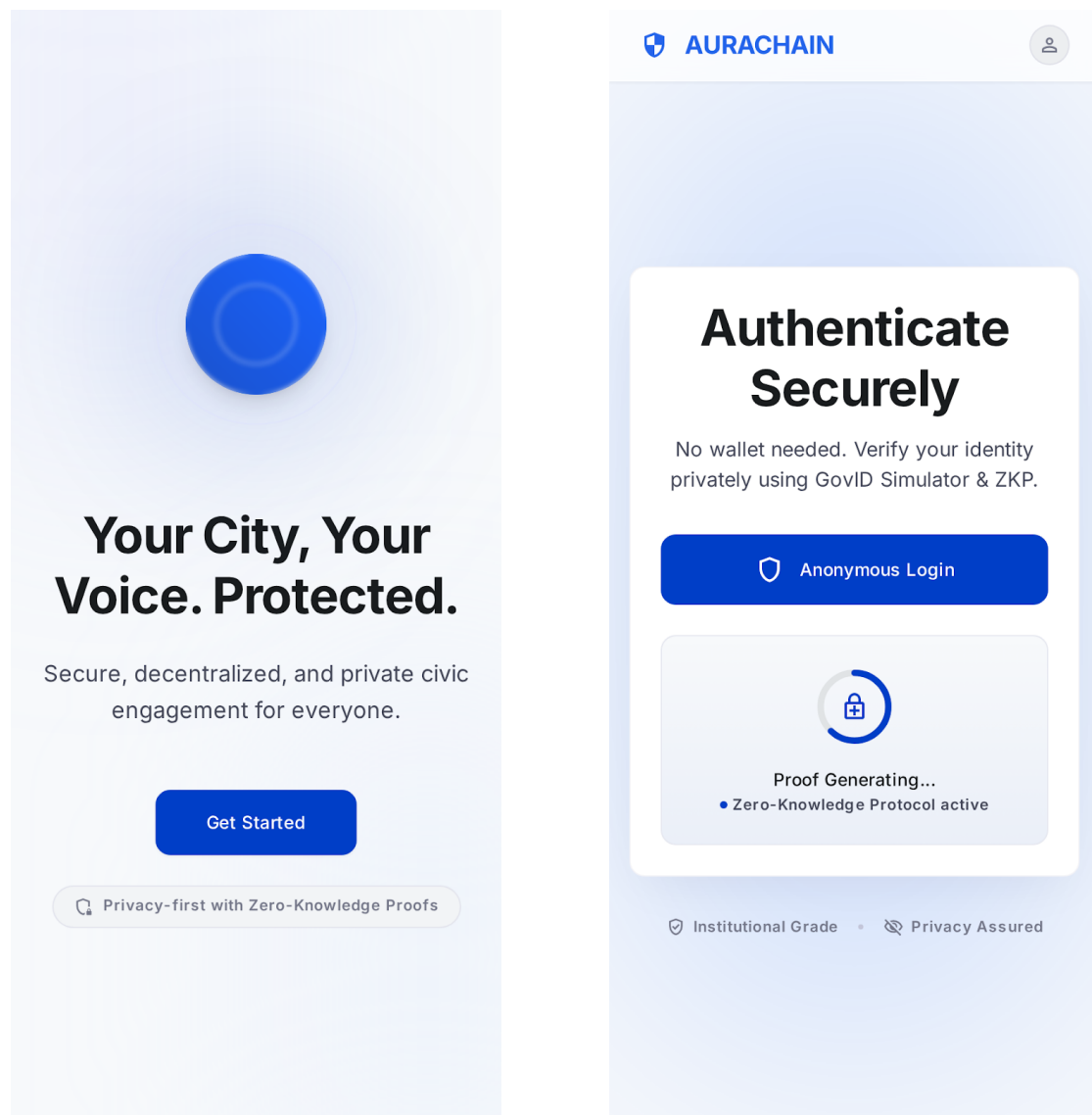


Figure 4.1: Citizen Authentication Interfaces

2. **Report Submission and Cryptographic Signing:** The authenticated user submitted a test report containing a text description (e.g., "Pothole on Main Street") and a sample image. The DApp successfully consumed a ZKP ticket, hashed the payload, and signed it using the local private key before transmitting the multipart form data to the backend relay.

 AURACHAIN

Create New Report

Securely submit an incident or observation to the public ledger.

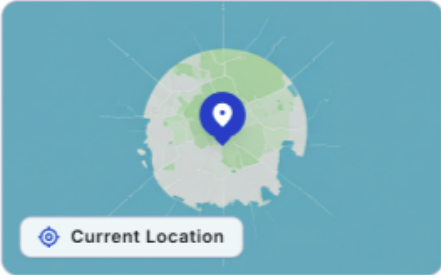
Step 1: What's the issue?

Describe the situation clearly and objectively...

Step 2: Add Proof


Upload Photo/Video
Supports JPG, PNG, MP4

Step 3: Location


Current Location

Submit Anonymously

Your identity is cryptographically protected.

Feed

Report

Profile

Notifications

Figure 4.2: Report Submission Form

3. **Off-chain Storage and AI Moderation Routing:** Upon receiving the payload, the backend relayer processed the test data. The visual evidence was routed to the IPFS integration module, successfully generating a Content Identifier (CID). Simultaneously, the text and image data were evaluated by the AI moderation service, which validated the report as legitimate, non-abusive content.
4. **Blockchain Recording via Meta-Transaction:** Following AI approval, the backend relayer formatted the data—including the IPFS CID, issue description, and the cryptographic proofs—and submitted a meta-transaction to the smart contract on the Geth PoA network. The transaction was successfully mined by the validator nodes, securely anchoring the report to the immutable ledger without requiring the citizen to pay gas fees.
5. **Frontend Verification:** Finally, the DApp’s query interface was refreshed to retrieve the latest blockchain state. The newly submitted report appeared in the community feed with its corresponding status badge, upvote count, and image thumbnail, confirming that the entire decentralized pipeline executed flawlessly.

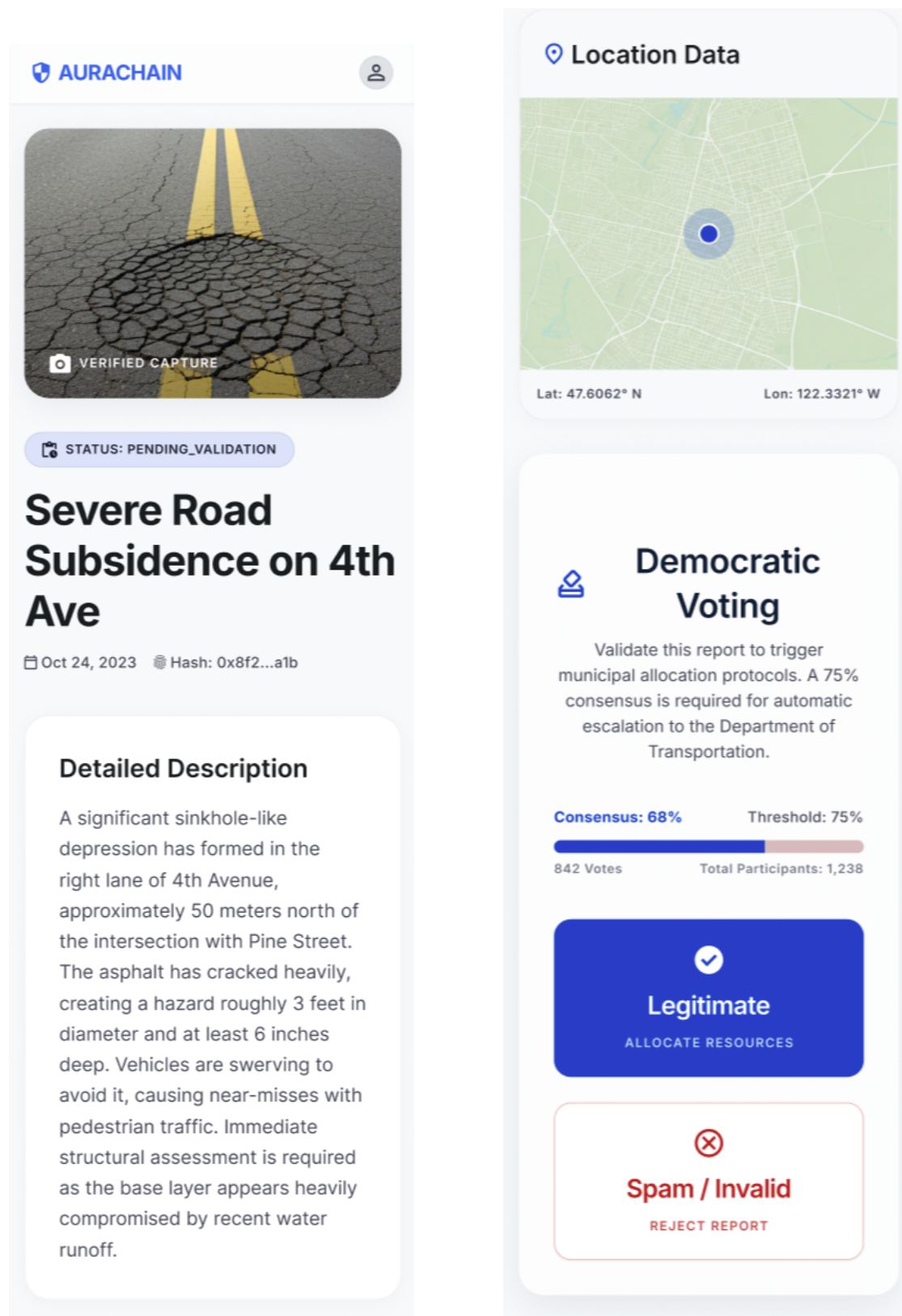


Figure 4.3: Views of the Submitted Report in the Community Feed

The successful execution of this simulated workflow validates the system architecture and confirms that the individual components interact correctly to provide a secure, anonymous, and user-friendly civic reporting platform.

4.8 Results and Observations

Use:

tables, graphs, screenshots.

Very important according to coordinator.

4.9 Comparison with Existing Systems

To demonstrate the advantages and novel contributions of the proposed community-assisted reporting framework, it is essential to compare its architecture and features against existing solutions in the domain of e-governance and citizen reporting.

4.9.1 Centralized Reporting Platforms

Traditional crowdsourced reporting systems, such as FixMyStreet and SeeClick-Fix, have successfully established user-friendly interfaces with intuitive reporting workflows and clear feedback loops. However, as identified in the literature, these platforms rely entirely on centralized server architectures. This centralization creates single points of failure and makes the systems susceptible to data tampering and unauthorized modifications. Furthermore, centralized platforms often require mandatory identity disclosure, which can discourage citizen participation, especially when reporting sensitive issues or criticizing local authorities. In contrast, the proposed system leverages a permissioned blockchain to ensure data immutability and absolute transparency. By employing Zero-Knowledge Proofs (ZKP) and identity decoupling, the proposed framework guarantees citizen anonymity while maintaining mathematical accountability, a feature completely absent in conventional centralized systems.

4.9.2 Traditional Content Moderation

Existing governance platforms typically rely on manual content moderation by human administrators to filter out spam or abusive content. While effective for small-scale operations, manual moderation introduces significant transaction latency, high operational costs, and the potential for human bias. Most importantly, it reintroduces centralized control over what public information is allowed to be published. The proposed system addresses these limitations by integrating

an AI-based off-chain moderation oracle. By automating the filtering of textual and visual content using machine learning models before the data is permanently recorded on-chain, the system achieves scalable, unbiased, and rapid moderation while preserving the decentralized ethos of the platform.

4.9.3 Public Blockchain Approaches

While some theoretical governance models propose using public, permissionless blockchains (such as the Ethereum mainnet) to achieve maximum decentralization, these architectures are poorly suited for frequent local governance interactions. Public blockchains suffer from high transaction latency, limited throughput, and unpredictable, fluctuating gas fees that the user must pay. The proposed framework overcomes these usability challenges by utilizing a private, permissioned blockchain running a Proof-of-Authority (PoA) consensus mechanism. This architecture provides predictable, high-performance transaction processing. Additionally, it eliminates the financial burden on citizens through the use of a backend relay network (meta-transactions), offering a seamless, "gasless" experience.

4.9.4 Domain-Specific Blockchain Solutions

Recent literature highlights several blockchain-based complaint management systems, but these are primarily tailored for highly specific domains such as law enforcement (e.g., police complaint registries). These systems often focus narrowly on administrative efficiency, restrict access entirely to authorized personnel, and frequently lack robust mechanisms for handling large multimedia evidence. Unlike these isolated solutions, the proposed framework introduces a generalized, community-assisted approach. It actively encourages public participation through opinion polling and community upvoting. Furthermore, it efficiently manages large multimedia files (e.g., images of infrastructure damage) through decentralized off-chain storage via IPFS, seamlessly linking cryptographic hashes back to the immutable blockchain ledger.

Chapter 5

Discussion

This chapter explains:
significance, novelty, limitations, insights.
Coordinator explicitly emphasized this.

5.1 Novel Contributions of the Project

Examples:
pseudonymous reporting, decentralized moderation, hybrid architecture.

5.2 Overall System Validity

Discuss:
whether objectives achieved, practical feasibility.

5.3 Advantages of Proposed Architecture

Examples:
transparency, modularity, scalability.

5.4 Limitations of Current Prototype

Examples:
simulated ZKP, limited oracle decentralization, controlled testing environment.

5.5 Real-world Applicability

Discuss:

local government usage, public reporting systems, governance transparency.

Chapter 6

Conclusion and Future Work

6.1 Conclusion

Summarize:

achievements, implementation progress, significance.

NOT same as abstract.

6.2 Future Work

Examples:

real ZKP integration, advanced AI models, distributed IPFS clusters, mobile application, DAO governance.

Appendix A

Smart Contract Code

Appendix B

Oracle API Endpoints

Appendix C

Docker Compose Files

Appendix D

System Diagrams

Appendix E

Testing Screenshots

Appendix F

GitHub Repository Links

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